Professor Dr. J. T. Andrews

1. Personal Information

(i)	Name	JOSEPH THOMAS ANDREWS
(ii)	Qualification	Ph.D. (Faculty of Engg.), M.Sc. (Appl. Phys., Faculty of Engg.) B.Sc. (Physics Honours)
(iii)	Designation	Professor, Head and Dean (SW)
(iv)	Email-id	jtandrews@sgsits.ac.in
(v)	Employee No.	1400 618
(Vi)	Department	Applied Physics & Optoelectronics
(vii)	Experience	26 Years

2. Educational Qualification

B.Sc. Physics (Hons)

1992 - MKU, Madurai

M.Sc. Appl. Phys. (Engg. Discipline)

1994 - DAVV, Indore

Ph.D. Appl. Phys. (Engg. Discipline)

1998 - DAVV, Indore

CMI-Level 5

2020 - Dudley,UK

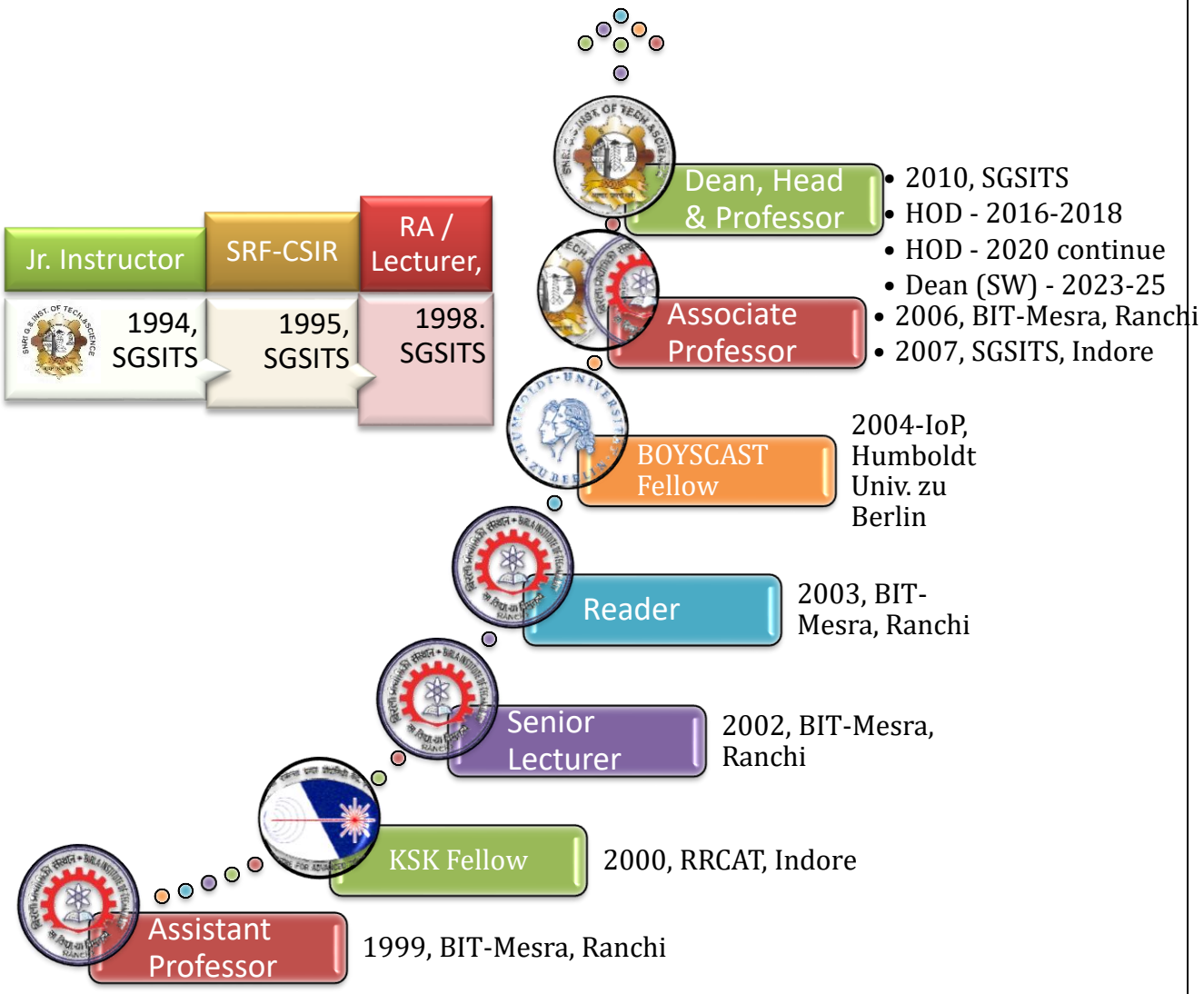
3. Research Interests

Optical Tweezers, Manipulation of nanomaterials, Biomedical Imaging, Non-invasive glucose measurements with Optical coherence tomography, Optical properties of Quantum dots, Quantum Computation Algorithm, Quantum Optical Computer, and NDT&E using real-time holography

4. Major Awards & Recognitions

- **Dr. K. S. Krishnan Award** by Department of Atomic Energy, Mumbai (2000)
- **BOYSCAST Award** of Department of Science & Technology, New Delhi (2002).
- **Young Scientist**, Department of Biotechnology under RGYST scheme (2007)
- Nominated speaker for **Dr. N. S. Satyamurthy Award** of Indian Physics Association.
- **Dr Khandelwal Memorial** Lecture during 35th Annual IAPT Convention (2021).

5. Career Graph



6. List of Enclosures

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Enclosure - 1: Academic & Administrative Responsibilities held

Academic / Teaching Experience and subjects taught

- **B.E./B.Tech.** – Physics (I year), Engineering Physics (I year), Material science (II year), Engineering Materials (II year), Digital Signal Processing (III year), Biomedical Instruments (IV year)
- **M.Sc.** - Solid state physics, Statistical mechanics, Lasers and applications, Mathematical methods, Computer programming, Nanoscience & Technology, Fiber & Integrated Optics, Engg. Drawing.
- **M.Tech.** - Lasers, Laser Material Processing, Applied Optics, Optical Instrumentation, Integrated Optics, Nanophotonics
- **Ph.D.** - Computational Methods

Other Academic

1. Dean Staff Welfare since 2023
2. Professor & Head since 2016
3. Professor since 2010
4. Associate Professor – SGSITS, Indore from Dec. 2007 – Mar. 2010
5. Assistant / Associate Professor – Birla Institute of Technology, Mesra, Ranchi (13, June 1999 – 13 Dec. 2007)
6. Senior Research Fellow of CSIR (1995-1998)
7. Junior Instructor – SGSITS, Indore (1994-95)
8. Visiting Research Fellow : Humboltd Universitat zu Berlin, Germany
9. Worked at Biomedical Applications Section, Raja Ramanna Center for Advanced Technology, Indore for 18 months.
10. As a BOYSCAST Fellow (Post Doctoral Research) worked at Instut fur Phyzik, Humboldt Universitat zu Berlin, Germany
11. Worked as Post Doctoral Fellow at SGSITS under National Laser Programme
12. Visited Hokkoido University, Japan for short term Research.

Administrative Experience

1. Dean-Staff Welfare (2023-2025)
2. Head of Department (2016-2025)
3. Warden – Transit Hostel (SGSITS, Indore), Boys Hostels (BIT Mesra)
4. Asst. Center Superintend of various exams including GATE, PET, JEE, etc.
5. Center Manager for various exams of UPSC/MPPSC/Mhow Cant Board/AGL, etc.
6. Coordinator of Exams for TOEFEL, GRE
7. Coordinator – National MEMS Design Center
8. In-Charge – Printing and Publishing (of Institute Annual Magazine, Newsletter, Exam

answer sheets, Mark sheets, Various documents, etc.)

9. Organised various Conferences and Workshops (Pl. see Annexure-IV)
10. Chairman – CWN committee- SGSITS
11. In charge of ERP/MIS implementation, MIS document preparation under TEQIP-P3
12. Nodal officer for AICTE, NIRF, AISHE, CSAB, GIAN
13. Nodal Officer – DTE Help Center and Kiosk (2008-2018)
14. Deputy Controller of Exams, SGSITS
15. Superintend of Exams for UG & PG.
16. Officer on Special Duty – Central Valuation.
17. Member of - Discipline committee, Exam Subcommittee, Purchase Committee, TEQIP project implementation committee, Admission Committee, Campus wide network committee, Academic council, etc.
18. In-charge - Physics Laboratory, -Language Laboratory, -Laser Laboratory, -Applied Photonics Laboratory
19. Coordinator of National Service Scheme (2 credit course)

Enclosure - 2: Research Experience & Doctoral Thesis Guided

Citations H-index: **11 (342+249 Citations)**

Google Scholar H-index: **13 (602 Citations)**

i-10 Index: **16**

Area of Research Interest:

Biomedical Imaging, Optical Coherence Tomography, Laser spectroscopy, Laser material processing, Digital holography, MEMS and MOEMS devices, FinFET devices, Quantum dots, Nonlinear Optics, Quantum Optics.

1. As a ***coordinator of National MEMS Design Center,*** taken MOEMS fabrications with IISc., Bangalore and IIT-Mumbai.
2. Completed 13 R&D sponsored projects and currently having three ongoing projects. Total value of projects exceeds 250 lacs.
3. As a ***BOYSCAST fellow*** worked with Prof. Henneberger, Director, Institut fur Physik, Humboldt Universitat zu Berlin, Berlin, Germany during Mar. 2004 – Mar. 2005.
4. As a ***Dr. K. S. Krishnan Fellow*** worked with Dr. P. K. Gupta, Head, Biomedical Applications Section, RRCAT, Indore during Nov. 2000 - Feb. 2002.
5. Awarded degree of doctor of philosophy and submitted a thesis entitled "A study of coherent radiation-matter interactions", under the supervision of Dr. (Mrs.) Pratima Sen in July 1998.
6. Post doctoral Research Associate (DAE-NLP), Aug. 1998 – July 1999, Dept. of Appl. Physics, Shri G S Inst. of Technology & Science, Indore
7. Senior Research Fellow (CSIR), Aug. 1996 – Aug. 1998, Dept. of Appl. Physics, Shri G S Inst. of Technology & Science, Indore
8. Carried out a theoretical project "Rabi oscillations in III-V semiconductors" for the award of M.Sc. (Appl. Phys.) under the supervision of Dr. (Mrs.) Pratima Sen, which has been published in Solid State Communication, 95, 117 (1995) (UK).
9. Successfully completed one month summer training on the preparation and study of a high Tc superconductors and thin films in the laboratories of Inter University Consortium for DAEF, Indore during May-June 1993.

Ph.D. Thesis (Guided / Awarded and Supervising)

1. Dr. K. Bose (Applied Physics): *Testing and Measurement of Physical Parameters using holographic interferometry* (2007), BIT, Mesra, Ranchi, JH.
2. Dr. R. Poddar (Biotechnology) : *Non-invasive pathological measurements using Optical Coherence Tomography* (2008), BIT, Mesra, Ranchi, JH.
3. Dr. (Mrs) S. R. Sharma (Electronics & Instrumentation): *Imaging of biological samples using Optical Coherence Tomography* (2009), BIT, Mesra, Ranchi, JH.
4. Dr. S. Chattopathyay (Physics): *Studies of Optical and Electronic Properties of Core/Shell Quantum Dots*, Submitted (2012), SoP, DAVV, Indore, MP.
5. Dr. J. Solanki (Physics/Medical Instrumentation) : *Non-invasive, Non-contact Monitoring of Blood Glucose using Optical Coherence Tomography*, (2013), SoP, DAVV/SGSITS, Indore, MP.

6. Dr. Om Prakash Choudhary, *Design, Fabrication and Characteriation of Optical Coherence Tomography on a Chip*, (Applied Physics) (2019), SGSITS/DAVV, Indore MP.
7. Mr Dheeraj Dhoolani: *A Study of Non-linear Optical Properties of Metamataerials*, (Physics) (2021), SoP, DAVV/SGSITS, Indore, MP
8. Shri Manish Mehta, *Quantum Computing with Semiconductor Quantum Dots*, (Physics) (2024), SoP, DAVV/SGSITS, Indore, MP
9. Shri Lokendra Chouhan (2025), *Study, Design, Optimization and Fabrication of Integrated Optical Low Coherence Reflectometer*, SoP, DAVV/SGSITS, Indore, MP
10. Ashish Singh Bais (2025), *Design optimization and fabrication of integrated optical coherence tomography on a single chip*, SoP, DAVV/SGSITS, Indore, MP
11. Surendra Rawat (2026), *Study of co-eventum Mechanics to Quantum Logic Operations*, SoP, DAVV/SGSITS, Indore, MP.

Enclosure - 3: R & D Projects Ongoing / Completed

S. No.	Name of the funding agency	Name of the Scheme	Programme Title	Year of Funding	Duration	Amount Sanctioned (Rs. lacs)	Status: Ongoing/ Completed
1.	AICTE	RPS	NDT of Thin materials using Holography	2000	2 years	6.00	Completed
2	DST	SERC	Optical prop. Of stratified media	2004	3 years	20.00	Completed
3	ICTP	TWAS	Optical prop using OCT	2004	1 year	Euro 3000	Completed
4	DST	BOYSCAST	Optical control in Quantum Dots	2005	1 year	30.00	Completed
5	Gmbh Res. grant		Optical Control in Quantum Dots	2006	6 months	Euro 45000	Completed
6	DRDO		Thermal Damage in tissue	2007	3 years	20.00	Returned due to transfer of place
7.	AICTE	RPS	Tissue imaging using OCT	2006	2 years	7.20	Completed
8.	DBT	RGYI	Frost burn in tissue using OCT	2007	3 years	22.00	Ongoing
9	MPCST	R&D	Blood glucose using OCT	2011	2 years	14.85	Completed
10	UGC	Major	Fabrication of PVC	2012	2 years	10.00	Completed
11	ICMR	--	Wearable Glucometer	2012	2 years	74.50	Completed
12	UGC	--	Nanomaterial	2015	3 years	12.00	Completed
13	AICTE	MODROBS	Laser and Fiber Optics	2016	2 years	11.81	Completed
14	DST	SERB	MOEMS glucometer	2020	3 years	33.02	Completed
15	AICTE	RPS	MOEMS OCT fabrication	2021	2 years	12.00	Completed
Completed Projects: 11 Ongoing Projects: 02				Total Money Generated: ₹ 2,44,88,000.00 Total Manpower Generated: Ph.D.- 8, PG - 45			

Enclosure - 4: Journal Publications

S. No	Title of the Research Paper	Name & *Journal*	**Vol. No.** & Page.	Year / Remarks
1.	Rabi oscillations in III-V semiconductors,	J. T. Andrews and P. Sen *Solid State Commun.,*	**90,** 117-120	1995
2.	Effect of magnetic field on Rabi oscillations in III-V semiconductors,	J. T. Andrews, P. Sen and P.K. Sen, *Nonlinear Optics,*	**12,** 327-337	1995
3.	Fiber Optical Communication,	J. T. Andrews, P. Sen and R. R. Puri Ed. A.K Sharma	**Viva, New Delhi**	1998 (Book)
4.	Computers and Devices for Communication	J. T. Andrews, P. Sen and P. K. Sen	**Allied, New Delhi**	1998 (Book)
5.	Theory of free induction decay,	J. T. Andrews, P. Sen and P. K. Sen *Ind. J. Pure and Applied Physics,*	**36,** 655-660	1998
6.	Effect of noncentrosymmetry on stark broadening in bulk Indium Antimonide crystal,	J. T. Andrews and P. Sen *J. Quantum and Semiclassical Optics,*	**10,** 663-669	1998
7.	Stark shift and self energy correction in noncentro-symmetric systems,	J. T. Andrews and P. Sen *Nonlinear Optics,*	**19,** 177-188	1998
8.	Ultrafast Phenomena in Semiconductors	J. T. Andrews and P. Sen *Proc.SPIE*	**3624,** 57-66	1999
9.	Absorption spectra of small semiconductor quantum dots,	J. T. Andrews and P. Sen *Superlattice and Nanostructures,*	**26,** 171-180	1999
10.	Optical nutation and free induction decay in magnetized narrow gap III-V semiconductors,	J. T. Andrews and P. Sen *J. Nonlinear Optical Phys. And Materials*	**8,** 431-441	1999
11.	Optical absorption spectra of a QD in a microcavity	J. T. Andrews, P. Sen and R. R. Puri *J. Physics B, Cond. Matter*	**11,** 6287-6300	1999
12.	Optical Pulse and Beam Propagation	J. T. Andrews and P. Sen *Proc. SPIE*	**3927,** 52-62	2000
13.	Nonlinear Materials, Devices, and Applications	J. T. Andrews and P. Sen *Proc.SPIE*	**3928,** 234-244	2000
14.	Laser Applications in Microelectronic and Optoelectronic Manufact.	J. T. Andrews and P. Sen *Proc.SPIE*	**3933,** 14-23	2000
15.	Effect of polarisation state on Squeezed state generation in GaAs	H. K. Gahir, P. Sen. J. T. Andrews and R. R. Puri *J. Quant. & Semiclassical Optics,*	**2,** 1-10	2000
16.	Steady state nonlinear gain in small quantum Dots	P. Sen and J. T. Andrews *Solid State Commun.*	**120,** 195-200	2001
17.	Pump-probe absorption	P. K. Sen and J. T.	**29,** 287-300	2001

	characteristics of small quantum dots	Andrews		
18.	Nonlinear Absorption in semiconductor quantum dots,	J. T. Andrews and P. Sen *Virtual J. Nanoscale Science & Technology,*	**Mar 11,** 05	2002
19.	Nonlinear Absorption in semiconductor quantum dots	J. T. Andrews and P. Sen *J. Appl. Physics*	**91,** 2877-2882	2002
20.	A novel scanning technique for Optical coherence tomography	J. T. Andrews, H. S. Patel and P. K. Gupta *J. Optics*	**30,** 151-157	2002
21.	Optical binding in a line trap	S. K. Mohanty, J. T. Andrews and P. K. Gupta *Optics Express*	**12,** 2746-2753	2004
22.	Quantum Optical Phenomena in Semiconductor Quantum Dots	J. T. Andrews *Quantum Dots: Research Developments,* Ed. P. A. Ling	**Nova, New York** 149-161	2005 (Book)
23.	Errors in Abel inversion Transformations	R. K. Paul, K. Bose, J. T. Andrews & P. K. Barhai *Plasma Devices & Applications*	**13,** 281-289	2005
23.	Measurement of scattering coefficient with blood glucose concentration using diffuse reflectance,	R. Poddar, S. R. Sharma, P. Sen & J. T. Andrews *J. Biochem. & Biotechnology*	**10,** 312-315	2006
24.	Digital real-time holography for fault detection	K. Bose, J. T. Andrews and P. K. Barhai *Int. J. Mat. Science*	**2,** 137-144	2006
25.	Understanding laser modes using Optical Coherence Tomography	R. Poddar, S. R. Sharma, K. Bose, P. Sen & J. T. Andrews *Am. J. Physics*	**74,** 569-571	2006
26.	Stimulated emission from the biexciton in a single quantum dot	I. A. Akimov, J. T. Andrews & F. Henneberger *Phys. Rev. Lett.*	**96,** 067401	2006
27.	Low cost experiments with laser and personal computer	J. T. Andrews, B. Gupt, K. Bose and A. Kumar *Transactions on Engineering Education*	**4,** 155-157	2007
28.	Study of Correlation Between Glucose concentration and Reduced Scattering Coefficients in Turbid media using Optical Coherence Tomography	R. Poddar, S. R. Sharma, P. Sen and J. T. Andrews *Current Science*	**95,** 340-346	2008
29.	Diagnosis of thermal damage in tissue using PS-OCT	S. R. Sharma, R. Poddar, P. Sen and J. T. Andrews *Afr. J. Biotechnology*	**7,** 2049-2054	2008
30.	Optical Coherence	R. Poddar, S. R. Sharma, P. Sen and J. T. Andrews	**Aug. 3**	2008

	Tomography	*Optical Coherence Tomography News,*		
31.	All optical quantum CNOT gate in semiconductor quantum dots	Md. S. Qureshi, P. Sen, J. T. Andrews and P. K. Sen *IEEE. J. Quantum Electronics*	**45,** 59-65	2009
32.	Effect of shell thickness on exciton and biexciton binding energies in ZnSe/ZnS core/shell quantum dot	S. Chattopadhyay, J. T. Andrews, P. Sen, P. K Sen, *IEEE-CODEC: Proc. 2009 4th Int. Conf. Comp. and Devices for Comm.,*	**ISBN:978-1-4244-5073-2**	2009
33.	Impact of shell thickness on exciton and biexciton binding energies of a ZnSe/ZnS core–shell quantum dot	P. Sen, S. Chattopadhyay, J. T. Andrews, P. K. Sen *Journal of Physics and Chemistry of Solids*	**71,** 1201-1205	2010
34.	Towards a Wearable Non-invasive Blood Glucose Monitoring Device	J. T. Andrews, J. Solanki, Om P Choudhary, S. Chouksey, N. Malvia, P. Chaturvedi and P. Sen. *J. Phys.: Conf. Ser.*	**365,** 012004	2012
35.	Effect of Shell and Shell Thickness on Photoluminescence (PL) of a CdSe/ZnS Core – Shell Quantum Dot	S. Chattopadhyay, P. Sen, J. T. Andrews, P. K. Sen *J. Phys.: Conf. Ser.*	**365,** 012037	2012
36.	Semiconductor core-shell quantum dot: A low temperature nano-sensor Material	S. Chattopadhyay, P. Sen, J. T. Andrews, P. K. Sen *J. Appl. Phys*	**111,** 034310	2012
37.	Blood glucose monitoring in human subjects using optical coherence tomography	J. Solanki, P. Sen, J. T. Andrews & K. K. Thareja *J. Optics*	**41,** 127-133	2012
38.	Cyclic Correlation of Diffuse Reflected Signal with Glucose Concentration and Scatterer Size	J. Solanki, P. Sen, J. T. Andrews & K. K. Thareja J. Modern Physics	**3**, 1745	2012
39.	Special Mention in News	OCT News	**Feb. 21**	2012
40	Polarization sensitive optical low-coherence reflectometry for blood glucose monitoring in human subjects	J. Solanki, O.P. Choudhary, P. Sen, J. T. Andrews *Review of Scientific Instruments*	**84**, 073114	2013
41	Trapping sub-micron sized particles in holographic optical tweezers,	G. Dwivedi, A. Gupta, M. Shukla, S. Kanaujia, S. Yede, J. T. Andrews, *J. Physics. CS*	**534**, 012066	2014
42	MOEMS Optical delay line for optical coherence tomography	O. P. Chaudhary, S. Chouksey, P. K. Sen, P. Sen, J. Solanki, J. T.	**534**, 012066	2014

		Andrews *J. Physics. CS*		
43	Low temperature nano-spin filtering using a diluted magnetic semiconductor core–shell quantum dot	S Chattopadhyay, P Sen, JT Andrews, PK Sen *Physica E: Low-dimensional Systems and Nanostructures*	**61,** 198-203	2014
44	Microelectronics Facilities at SGSITS, Indore	J. T. Andrews *Sc. Res. in Appl. Sciences,*	**3**, 27-32	2015
45	Effect of oxygen partial pressure on the structural and optical properties of ion beam sputtered TiO2 thin films	FA Tantray, A Agrawal, M Gupta, JT Andrews, P Sen *Thin Solid Films*	**619,** 86-90	2016
46	Negative thermo-optic coefficients and optical limiting response in pulsed laser deposited Mg-doped ZnO thin films	A Agrawal, TA Dar, JT Andrews, PK Sen, P Sen *J. Opt. Society of America-B*	**33,** 2015-19	2016
47	Role of Fermi energy on the properties of split ring resonators	Dheeraj Dulhani, Pratima Sen, Joseph Thomas Andrews AIP: Emerging Interfaces of Phys. Sc.	2224:060001 DOI: 10.1063/ 5.0000790	2020
48	Room temperature CNOT operation in a diluted magnetic semiconductor quantum dot	Manish Kumar Mehta, Joseph Thomas Andrews, Pratima Sen Quant. Inf. Proc.	**20**, 194 DOI: 10.1007/ s11128-021-03100-5	2021
49	All Optical Integrated MOEMS Optical Coherence Tomography System	Ashish Singh Bais, Lokendra Singh Chouhan, Joseph Thomas Andrews, J. Physics	**2426,** 01204 **http://dx.doi.org/ 10.1088/1742-6596/2426/1/012024**	2023
50	A Novel Chirped Bragg Grating Based Spectral Domain Delay Generator	Lokendra Singh Chouhan, Ashish Singh Bais, Joseph Thomas Andrews, J. Physics	**2426,** 01205 **http://dx.doi.org/ 10.1088/1742-6596/2426/1/012025**	2023

Enclosure - 5: Patents Publications/Sanctioned

1. Thomas Andrews Joseph, Ashish Singh Bais and Lokendra Singh Chouhan, A Controllable Optical Delay Line Generator and Method of Operation Thereof, Appl. No.202421074742 A, 03/10/2024, Indian patent (Published).
2. Thomas Andrews Joseph, Lokendra Singh Chouhan and Ashish Singh Bais, Chirped Bragg grating based spectral domain optical coherence tomography device, Appl. No. 202521005548 A, 23/01/2025, Indian patent (Published).

Enclosure - 6: Selected Publications in Conferences

1. **J. Thomas Andrews**, P. Sen and P. K. Sen (1996) Rabi oscillations in magnetized Indium Antimonoide, *International Conf. Spectroscopy: Perspectives & Frontiers (INCONS), BARC, Bombay*, p. 143.
2. **J. Thomas Andrews**, P. Sen and R. R. Puri (1998), Absorption spectrum of a quantum dot, *Proc. International Conf. on Computers and Devices for Communication (CODEC-98) Calcutta,* pp. 567-570.
3. **J. Thomas Andrews** and P. Sen (1998) Absorption spectra of small semiconductor quantum dots, *Proc. Int. Conf. on Fibre Optics, Photonics'98, IIT-New Delhi* pp. 561-564.
4. **J. Thomas Andrews** and P. Sen (1999) Absorption spectra of small semiconductor quantum dots, *Photonics West - '99, San Francisco.*
5. **J. Thomas Andrews**, MNV Ravikumar, P. Sen and P. K. Sen (1999) Measurement of molecular weight of colloids using laser light scattering, *ETOP, Mexico*
6. P. Sen, **J. Thomas Andrews** and P. K. Sen (1999) Measurement of liquid flow using laser Doppler velocimetry, *ETOP, Mexico*
7. **J. Thomas Andrews** and P. Sen (1999) Dynamical Stark effect in small quantum dots, *Photonics West 2000, San Jose.*
8. **J. Thomas Andrews** and P. K. Sen (1999) Pump-probe absorption spectra of small quantum dots, *Photonics West - 2000, San Francisco.*
9. **J. Thomas Andrews** (2001) Dynamic Stark effect in Quantum dots, QDS-2000, Japan.
10. **J. Thomas Andrews**, H. S. Patel and P. K. Gupta (2001) A novel scanning technique for OCT, *Optonics 2001, Trivanathapuram.*
11. **J. Thomas Andrews**, H. S. Patel and P. K. Gupta (2001) Measurement of optical properties using optical coherence tomography, *DAE-BRNS National Laser Symposium-2001*, CAT, Indore.
12. R. Tiwari, P. Sen, **J. T. Andrews,** R. Jain, P. K. Dutta and P. K. Sen, Second order generation in conjugated polymers, *DAE-BRNS National Laser Symposium-2001*, CAT, Indore.
13. S. K. Mohanty, **J. Thomas Andrews** and P. K. Gupta (2002) Optical binding between dielectric Raleigh particles, *Proc. Int. Conf. Photonics – 2002*, TIFR, Mumbai, Dec. 16-18.
14. P. Sen and **J. Thomas Andrews**, Transient phenomena in semiconductor quantum dots, *Proc. Int. Conf. Photonics – 2002*, TIFR, Mumbai, Dec. 16-18.
15. **J. Thomas Andrews**, R. K. Paul, K. Bose and P. K. Barhai, Measurement of plasma parameters using optical method, *Proc. 17th National Symp. Plasma Sc. & Technology, Plasma – 2002*, Bharathiar University, Coimbatore, Dec. 16-19.
16. R. K. Paul, **J. Thomas Andrews,** K. Bose and P. K. Barhai, Development of Abel inversion program for the study of thermal plasma generated by arc generator, *Proc. 17th National Symp. Plasma Sc. & Technology, Plasma – 2002*, Bharathiar University, Coimbatore, Dec. 16-19.
17. R. Poddar, S. R. Sharma, P. Sen and **J. T. Andrews**, Effect of Glucose Concentration on Scattering Coefficients in Turbid Media Using Diffuse Reflectance, *Proc. Photonics 2004, BIO-14, CUSAT, Cohcin, India, Dec. 09-11, 2004.*
18. K. Bose, **J. T. Andrews** and P. K. Barhai, Material testing and Analysis Using Digital Real-Time Holography, *Proc. Photonics 2004, DISP-07, CUSAT, Cohcin, India, Dec. 09-11, 2004.* R. Poddar, S. R. Sharma, K. Bose, P. Sen and **J. T. Andrews**, A Study of Longitudinal Laser Modes Using Optical Coherence Tomography. *Proc. of V DAE National Laser Symposium, VIT, Vellore, Dec. 7-10, 2005.*
19. **J. T. Andrews**, I. A. Akimov and F. Henneberger, Coherent-control using stimulated emission from a biexcition in a single quantum dot, *Proc. of V DAE National Laser Symposium, VIT, Vellore, Dec. 7-10, 2005.*
20. K. Bose, R. K. Paul, P. K. Barhai and **J. T. Andrews**, Plasma diagnostics using digital interferometry technique, *Proc. of V DAE National Laser Symposium, VIT, Vellore, Dec. 7-10, 2005.*

21. R. K. Paul, **J. T. Andrews**, K. Bose and P. K. Barhai, Effect of equipment resolution on Abel Inversion, *Proc. Plasma 2005, CUSAT, Cochin, Dec. 5-7, 2005.*
22. K. Bose, R. K. Paul, P. K. Barhai and **J. T. Andrews**, Determination of Plasma parameters using digital real-time holography, *Proc. Plasma 2005, CUSAT, Cochin, Dec. 5-7, 2005.*
23. S. R. Sharma, R. Poddar, P. Sen and **J. T. Andrews**, Diagnosis of burn depth in skin using Polarisation Sensitive Optical Coherence Tomography, *Proc. SPIE, Photonics West, Feb (2006).*
24. S. R. Sharma, R. Poddar, P. Sen and **J. T. Andrews**, Diagnosis of burn depth in skin using Polarisation Sensitive Optical Coherence Tomography, *Proc. SPIE, Photonics West, Feb (2006).*

PUBLICATIONS IN CONFERENCES ARE NOT DOCUMENT SINCE 2007

Enclosure - 7: Workshops / Seminars Organised

1. Convener - First Colloquium on R & D projects at BIT, Mesra, Oct. 2002.
2. Convener -18th National Symposium of Plasma Science Society – Plasma – 2003, Dec. 2003.
3. Convener - World Year of Physics – 2005 with DST, Jharkhand, Nov. 2005.
4. Convener - TEQIP workshop on Academic Reforms, Dec. 2005.
5. Convener – XIV Regional meeting of DST councils and departments (Eastern Region) Feb. 2007.
6. Coordinator – Hindi Diwas Celebration with Dept. Atomic Energy, October 2008.
7. Coordinator – Three day National Science Day, Celebrations with support from MPCST, Feb. 2009.
8. Coordinator – Golden Jubilee year of Invention of Laser – Mar. 2010.
9. Co-convener - National Seminar on Innovations, Opportunitites & Challenges in Technical Education Mar. 2012.
10. One day workshop on COMSOL, Sep. 2012.
11. Convener – National Science Day Celebrations, Feb. - 2013.
12. Convener – National Workshop on Nanophotonics Sep. 2013.
13. Convenor – National Seminar on Optoelectronics, June 2015.
14. Coordinator – Two Weeks ISTE STTP on Engineering Physics, Dec. 08-18, 2015
15. Convenor - One Day Symposium on *"Moral Practices & Ethical Issues in Industrial Organizations and Academic Institutions*", March 16, 2015.
16. Convenor – Hands-on Training on "LaTeX for Academic Activities", March 25-26, 2017.
17. Convenor – Symposium on Modern Trends in Physics: A tribute to Prof. P. K. Sen, November 28, 2017.
18. Coordinator – Natural Learning Process, A two lecture series for School Children, Aug. 25-26, 2018.
19. Chairman – Two days STTP on "Lasers & Optical Communication", April 09-10, 2018.
20. Coordinator – 35th MP Young Science Congress, Feb 28-29, 2020.
21. Convenor – ATAL FDP on Sensors, July 2021.
22. Convenor -IPR awareness program under the flagship of Kalam Program for Intellectual Property Literacy and Awareness Campaign (KAPILA), June 04, 2022.
23. Coordinator - Ajadi ka Amrit Mahotsav and DAE Iconic Week Celebrations, August 25, 2022.
24. National Science Day Celebrations 2024 (Feb.28 – Mar 02, 2024)
25. Lectures on Innovation from Laboratory to Society, April 05, 2024.
26. National Science Day Celebrations 2025 (Feb. 28 – Mar 01, 2025).

Enclosure - 8: Testing Consultancy Work Completed

1. Specific Gravity Measurements of PVC pipe samples, Department of Water Resources, Government of Jharkhand.
2. Measurement of transmission spectra of Epoxy-Glass composites, Various Small Scale Industries at SIDCO, Ranchi.
3. Testing of optical plates / mirrors for surface quality, Optomechanical, Indore, (Light Guide Optics, Indore, Indore, DRDO-Dehradun, DRDE-Hyderabad)
4. GSM Testing (Mastermind Printers, Indore, Sainath Printeres, Indore, Sriram Printers, Indore, SGSITS, Indore)
5. UV Photoreactor Design (Kriti Soya, Indore)
6. UV Photoreactor Testing (Kriti Soya, Indore)
7. UV-C test of Lecithin (Kriti Soya Indore)
8. Stereographic studies of Dental Work (Modern Dental Hospital, Indore)

Enclosure - 9: Invited Lectures / Session Chair/ Key Note Lectures Delivered

1. Invited Speaker at "National Technology Day-2002" at Birla Institute of Technology, Mesra, during May 2002.
2. Invited Speaker at "21st National Symposium on Plasma Science & Technology, MNIT, Jaipur" during December 2006.
3. Invited Speaker at "IEE workshop on Nanomaterials and Nanotechnology" at Ranchi during November 2006.
4. Invited Speaker at " Recent Trends in Nanotechnology" at SGSITS, Indore during March 2007.
5. Resource Person at "BMI, SGSITS, Indore" during May 2009.
6. Invites Speaker at "Recent Development in Engineering Materials" at Birla Institute of Technology, Mesra, Ranchi during, May 12-14, 2011.
7. Resource Person at "Staff Development Program" at Sagar Institute of Research & Technology, Bhopal, June 20, 2011.
8. Invited Speaker at "National Conference on Applied Physics" at Lakshmi Narayan Institute of Technology, Indore July 18, 2011.
9. Invited Speaker, "International Conference on Recent Trends in Physics ICRTP-2012" at School of Physics, DAVV, Indore during Feb. 04-05, 2012.
10. Resource Person, "FDP on Nanoelectronics" at Swami Vivekanand College of Engineering, Indore during November 2012.
11. Resource Person, "STTP on Applied Science" at Lakshmi Narayan College of Technology, Bhopal during, April 2013.
12. Invited Speaker, "Emerging trends in Nanoscience and Nanotechnology" SGGSIET, Nanded during August 09, 2013.
13. Invited Speaker, "Recent Trends in Communication Network Technologies", SGSITS, Indore, Mar. 19, 2014.
14. Invited Speaker, "Bio-nanotechnology Workshop", Maharaja Ranjit Singh College of Professional Science, Indore, April 27-28, 2015.
15. Invited Speaker, "National Conference on Emerging trends in Nanoscience-ETNS", Medicaps Institute Technology & Management, Indore, Sept. 12, 2014.
16. Invited Speaker, "Seminar on New trends in Physics", Shri Sitram Jaju Govt Degree College, Neemuch, April 03, 2015.
17. Invited Speaker, "Faculty Development Program", Mata Jijabai Government Snatkottar Girls College, Indore, February 11, 2017.
18. Resource Person, "Ph.D. Students Lectures on Modern Physics", SOP, DAVV, Indore Feb. 02, 2018.
19. Resource Person, "Ph.D. Students Lectures on Modern Physics", SOP, DAVV, Indore March 07-08, 2019.
20. Invited Speaker & Resource Person, "Refresher Course on Nano-Physics and Electronics", SOP, Devi Ahilya University, Indore, Aug. 30-31, 2019.
21. Invited Speaker, "Modern Trends in Physics – UG Students", St Paul Inst. Prof. Studies, Indore, Oct. 12, 2019.
22. Invited Speaker, "Join conference of Indian Society for Oncology and Indian Society of Medical and Pediatric Oncology, ISO-ISMPO", RRCAT, Indore Dec. 2-3, 2019.
23. Keynote Speaker, "National Conference on ICT tools for Research", Govt. Girls PG College, Ujjain Feb. 14, 2020.
24. Invited Speaker, "NWADBP-2020-Applications of Physics to daily life" Swami Vivekanand Government College, Susner, July 18-19, 2020.
25. Resource Person, "International e-Symposium on Modern Trends in Physics & Biophotonics", Govt. College, Manawar, Dhar, June 6-10, 2020.
26. Resource Person, "e-Faculty Development Program", Shri Vaishnav Vidhyapeth University, Indore May, 4-8, 2020.
27. Invited Speaker & Resource Person, "Lecture Series at MSIT, MATS University", Raipur, April 23-26, 2020.

28. Plenary Speaker, “35th MP Young Scientist Congress”, SGSITS, Indore, Feb. 29, 2021.
29. Invited Speaker, “UKIERI-AICTE Technical Teachers Leadership Program”, Dudley College, UK, October 18, 2020.
30. Invited Speaker, “Synchrotron Users Symposium”, at RRCAT, Indore during Dec. 05, 2013.
31. Resource Person, “AICTE Atal FDP”, Inst. Engg. Technology, Devi Ahilya University, Indore, Oct. 20-24, 2020.
32. Invited Speaker – “National Mathematics Day”, Jagran Lakecity University, Bhopal, Dec. 23, 2021.
33. Invited Speaker – 29th National Laser Symposium, Tutorial Series, Shri Vaishnav Vidhyapeeth University, Indore, Feb. 10-11, 2021.
34. Judge and Session Chair – National Laser Symposium -NLS-29, Feb. 12-14, 2021.
35. Invited Speaker – National Science Day 2021, Acropolis Institute of Management and Research, Indore Feb. 28, 2021.
36. Resource Person, Webinar on “Quantum Computing and Quantum Logics” at Shri Vaishnav Institute of Management, Indore, June 01, 2021.
37. Resource Person, ATAL Faculty Development Program on Webinar on “Nonlinear Optical Sensors” at Shri G S Inst. Tech. & Science, Indore, July 07, 2021.
38. Invited Speaker, Orientation Program for I Year students, Jargan Lakecity University, Bhopal, Sep. 22, 2021.
39. Dr Khandelwal Memorial Lecture "Modern Physics Tools & Teaching Applied to Traditional Medicine Diagnostics" at 35th Annual IAPT Convention 2020-21 Shri Vaishanav Vidyapeth Vishwavidyalaya, Indore and Indian Association for Physics Teachers, Nov. 11-14, 2021
40. Keynote Speaker, "International Congress SANMANTRANNA - 2022", at Shri Vaishnav Vidhyapeeth University, Indore, Jan. 17-19, 2022.
41. Invited Speaker, "International conference on emerging techniques and functional materials ICETFM-2022" at Medi-Caps University, Indore, July 12-14, 2022.
42. Resource Person, "Refresher Course in Physics (Core)" at UGC-HRDC, Devi Ahilya Vishwavidyalaya, Indore, September 9-21, 2022.
43. Resource Person, "Tools & Techniques for Academic Writing, AIU-FDP" at Shri Vaishnav Vidyapeeth Univ., Indore, March 17, 2023.
44. Keynote Speaker, "National Science Day - 2024" Photonics Technologies for Vikshit Bharatat SICA College, Indore, February 28, 2024.
45. Invited Speaker, "National Science Day -2024" at SGSITS, Indore, March 15, 2024.
46. Keynote Speaker, "AVINYA 2024" Military College of Telecom Engineering, Mhow, September 27, 2024.
47. Invited Speaker, IDF Lecture Series 2.0, SGSITS, Jan 31, 2025

Enclosure - 10: Membership of Professional Bodies

1. Indian Physics Association (IPA – 11908).
2. Plasma Science Society of India (PSSI – 0469).
3. Indian Society for Smart Structures (ISSS - 404).
4. Institute of Electrical and Electronics Engineers (IEEE - 90861393).
5. Indian Society for Technical Education (Ranchi Chapter) (ISTE – 59021).
6. Indian Association for Physics Teachers (L-6652).
7. Indian Laser Association (ILA-LM-1441)
8. Optical Society of India (OSI)
9. Vigyan Bharati (VB-MVPS-10343)
10. Optical Society of America (OSA-976045)
11. Division of Plasma Physics, the Association of Asia Pacific Physical Societies (AAPPS-1174)
12. American Physics Society-Division of Laser Science (APS-62062636)

Enclosure - 11: Any other Information

Contributing regularly to support Open Source Software and actively participate in the following:

1. TeX user Group of India (TUG) and answers queries of new users of LaTeX.
2. Support openSUSE by developing Karamba widgets and a tester for various software of openSUSE.

   Supporting Sourceforge with development of few minor projects.

Registered User of the following

1. Nanotechnology Facilities Indian Nanoelectronics User Program at both IIT-Mumbai and IISc, Bangalore

   Center of Excellence in Nanoelectronics at IIT-Mumbai

Developed LaTeX templates for PG and Ph.D. thesis for use by the students and research scholars of Dept. Applied Physics. The templates are being by students during last five years. The templates are also being used by students of Holkar Science College, School of Physics, DAVV, etc.

Extra-Curricular Activities : Participated and obtained many prizes in Zonal / University level competitions on Pencil Sketching, Poster Coloring, Collage, Wax Carving and Folk Dances of Tamil Nadu (Oyilattam & Deverattam).

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